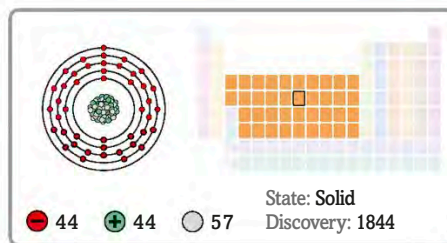


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# Ruthenium



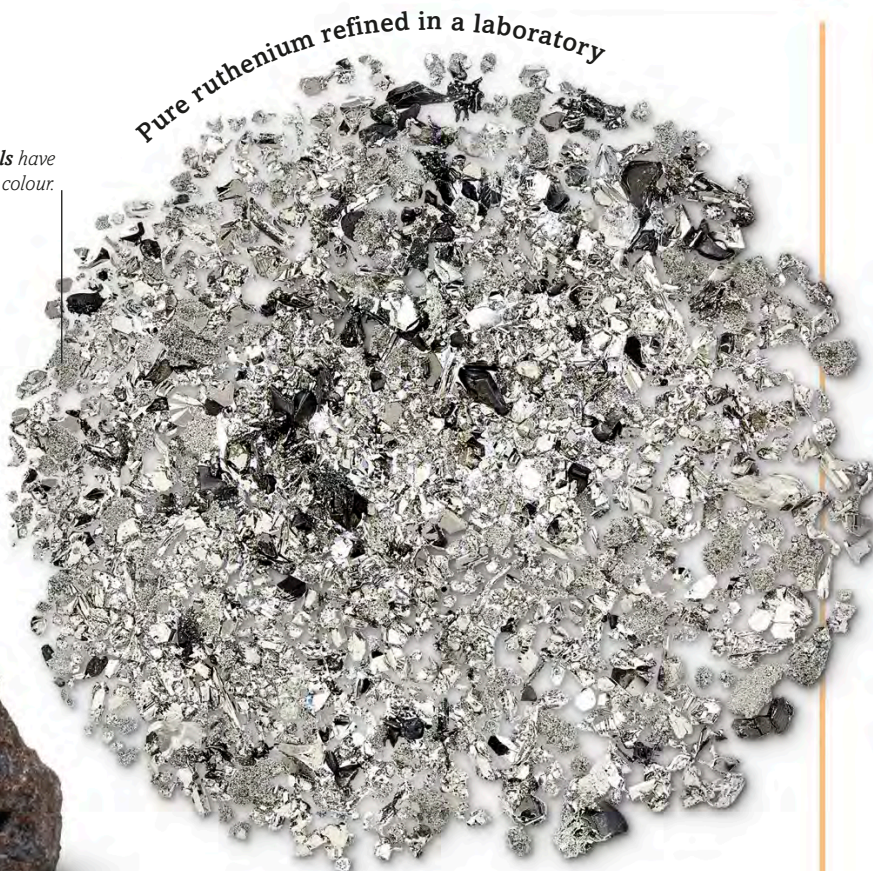
## Forms



Pentlandite

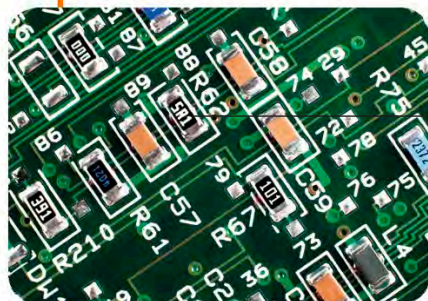
*This yellow-brown mineral is often found deep underground.*

*These crystals have a bright, silver colour.*



Pure ruthenium refined in a laboratory

## Uses

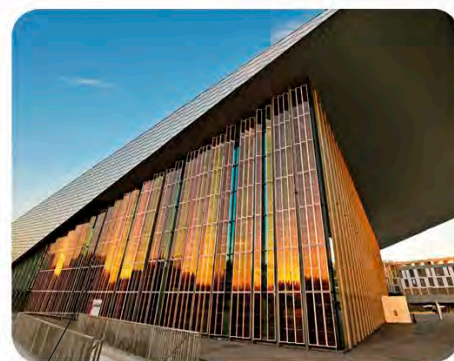


Electronic circuit board

*This component contains ruthenium dioxide.*



*The metal alloy in the switch is toughened by adding ruthenium.*



SwissTech Convention Centre, Switzerland

*These low-cost solar panels are made using ruthenium.*

Ruthenium is named after *Ruthenia*, an old Latin name for Russia. This rare metal is found in the mineral **pentlandite**, and its **pure form** is commonly extracted from this ore. A compound called ruthenium dioxide is used in several components in **electronic**

**circuits**, including resistors and microchips for computers and other digital devices. Adding a small amount of ruthenium makes softer metals, such as platinum and palladium, much tougher. Moving parts in devices such as **switches** benefit from this property.